

SOLUTIONS

MA 301
Test 1
Spring - 2013

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
6		10
7		10
8		10
Score		80

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1 : Since C is closed, let's try Green's Thm

$$N_x = 1$$

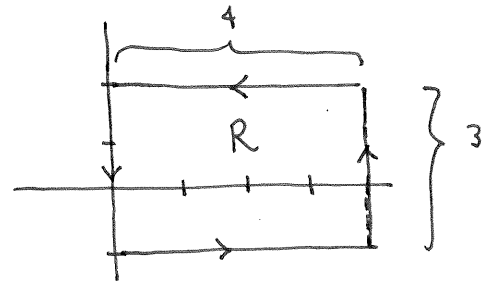
$$M_y = 4$$

$$\int_C \vec{F} \cdot d\vec{r} = \iint_R -3 \, dA$$

$$= -3 \cdot \text{Area}(R)$$

$$= -3 \cdot 12$$

$$= -36$$



Problem 2: The only thing we know about C is its initial and end points. Therefore, we can only solve this problem if the line integral is path-independent.

Lets show that this is the case by finding a $f(x,y,z)$ such that

$$\vec{F} = \nabla f. \implies \left[\frac{y^2}{x}, 2y \ln x, 3z^2 \right] = [f_x, f_y, f_z]$$

$$f_x = \frac{y^2}{x}$$

$$f = y^2 \ln x + g(y, z)$$

$$2y \ln x = f_y = 2y \ln x + g_y$$

$$g_y = 0$$

$$g = h(z)$$

$$f = y^2 \ln x + h(z)$$

$$3z^2 = f_z = 0 + h_z$$

$$h_z = 3z^2$$

$$h = z^3 + C \leftarrow \text{Let } C=0 \text{ since it will not change the fact that } \vec{F} = \nabla f$$

$$\implies f(x,y,z) = y^2 \ln x + z^3$$

This shows $\vec{F}$ is conservative and so $\int_C \vec{F} \cdot d\vec{r}$ is path-independent

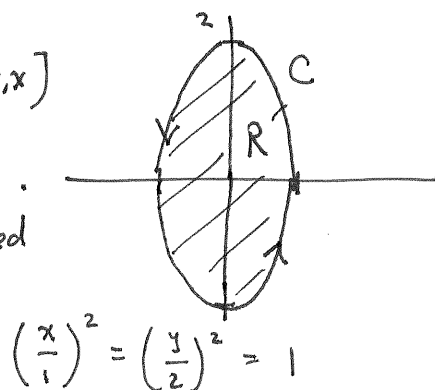
$$\begin{aligned} \text{So } \int_C \vec{F} \cdot d\vec{r} &= \int_{(2,0,1)}^{(1,1,4)} \nabla f \cdot d\vec{r} = f(1,1,4) - f(2,0,1) \\ &= (1)^2 \ln(1) + 4^3 - (0)^2 \ln(2) - 1^3 \\ &= 64 - 1 = 63 \end{aligned}$$

Problem 3 : (a) $\vec{F} = [-y, x]$ $N_x = 1$ $M_y = -1$

$$\frac{1}{2} \oint_C \vec{F} \cdot d\vec{r} = \frac{1}{2} \iint_R 1+1 \, dA = \iint_R dA = \text{Area}(R) \quad \checkmark$$

(b) $\text{Area}(R) = \frac{1}{2} \oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = [-y, x]$

To compute this line integral we need to parameterize C



$$\begin{aligned} x &= 1 \cos t \\ y &= 2 \sin t \end{aligned} \quad 0 \leq t \leq 2\pi$$

$$\Rightarrow \vec{r}(t) = [\cos t, 2 \sin t] \quad \Rightarrow \quad \vec{r}'(t) = [-\sin t, 2 \cos t]$$
$$F(\vec{r}(t)) = [-2 \sin t, \cos t]$$

$$\Rightarrow \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = 2 \sin^2 t + 2 \cos^2 t = 2$$

So

$$\begin{aligned} \text{Area}(R) &= \frac{1}{2} \oint_C \vec{F} \cdot d\vec{r} = \frac{1}{2} \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt \\ &= \frac{1}{2} \int_0^{2\pi} 2 \, dt = 2\pi \end{aligned}$$

Problem 4 : $\vec{F} = \rho \vec{\nabla} = [3z^2, 6, 6xz]$

$$\begin{aligned} x &= u \\ y &= u^2 \\ z &= v \end{aligned} \quad \Rightarrow \quad \vec{r}(u, v) = [u, u^2, v] \quad \begin{aligned} 0 &\leq u \leq 2 \\ 0 &\leq v \leq 3 \end{aligned}$$

$$\vec{r}_u = [1, 2u, 0]$$

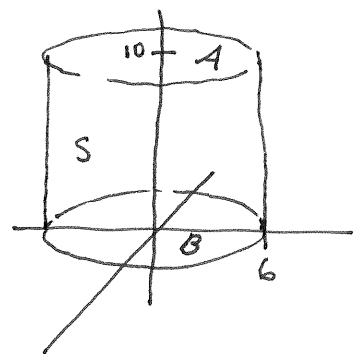
$$\vec{r}_v = [0, 0, 1]$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = [2u, -1, 0] \quad F(\vec{r}(u, v)) = [3v^2, 6, 6uv]$$

$$\begin{aligned} \text{Flux} &= \iint_S \vec{F} \cdot d\vec{A} = \iint_R \vec{F}(\vec{r}(u, v)) \cdot \vec{N} \, du \, dv \\ &= \int_0^3 \int_0^2 6uv^2 - 6 \, du \, dv \end{aligned}$$

Problem 5 : Let $S' = A \cup B \cup S$ and T the solid w/ boundary S'

$$\iint_{S'} \vec{F} \cdot d\vec{A} = \iint_A \vec{F} \cdot d\vec{A} + \iint_B \vec{F} \cdot d\vec{A} + \underbrace{\iint_S \vec{F} \cdot d\vec{A}}_{\text{Want this!}}$$



Note: $\text{div } \vec{F} = 7 - 2 + 1 = 6$

Since S is closed, we know (by the Divergence Thm)

$$\begin{aligned} \iint_{S'} \vec{F} \cdot d\vec{A} &= \iiint_T \text{div } \vec{F} \, dV = \iiint_T 6 \, dV = 6 \text{ Volume}(T) \\ &= 6 (\pi 6^2 \cdot 10) \\ &= 2160\pi \end{aligned}$$

For A we know: $z=10$ and $\vec{n} = [0, 0, 1]$ $\vec{F} \cdot \vec{n} = [7x+1, -2y, z] \cdot [0, 0, 1] = z = 10$

$$\begin{aligned} \text{So } \iint_A \vec{F} \cdot d\vec{A} &= \iint_A \vec{F} \cdot \vec{n} \, dA = \iint_A z \, dA \\ &= \iint_A 10 \, dA = 10 \text{ Area}(A) \\ &= 10 (\pi 6^2) \end{aligned}$$

For B we know: $z=0$ and $\vec{n} = [0, 0, -1] \Rightarrow \vec{F} \cdot \vec{n} = -z = 0$

$$\text{So } \iint_B \vec{F} \cdot d\vec{A} = \iint_B \vec{F} \cdot \vec{n} \, dA = 0$$

Therefore

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{A} &= \iint_{S'} \vec{F} \cdot d\vec{A} - \iint_A \vec{F} \cdot d\vec{A} - \iint_B \vec{F} \cdot d\vec{A} = 2160\pi - 360\pi - 0 \\ &= 1800\pi \end{aligned}$$

Problem 6 :

(a) S is not closed. Therefore we must use the direct approach.

Choose spherical coordinates with $\rho = 5$

$$\begin{array}{lll} x = 5 \sin \phi \cos \theta & u = \phi & \\ y = 5 \sin \phi \sin \theta & \longrightarrow & \vec{r}(u, v) = [5 \sin u \cos v, 5 \sin u \sin v, 5 \cos u] \\ z = 5 \cos \phi & v = \theta & 0 \leq u \leq \pi \quad 0 \leq v \leq 2\pi \end{array}$$

$$\vec{r}_u = [5 \cos u \cos v, 5 \cos u \sin v, -5 \sin u]$$

$$\vec{r}_v = [-5 \sin u \sin v, 5 \sin u \cos v, 0]$$

$$\vec{N} = |\vec{r}_u \times \vec{r}_v| = [25 \sin^2 u \cos v, 25 \sin^2 u \sin v, 25 \sin u \cos u]$$

$$\vec{F}(\vec{r}(u, v)) = [5 \sin u \cos v, 5 \sin u \sin v, 5 \cos u]$$

$$\begin{aligned} \vec{F} \cdot \vec{N} &= 125 \sin^3 u \cos^2 v + 125 \sin^3 u \sin^2 v + 125 \sin u \cos^2 u \\ &= 125 \sin^3 u + 125 \sin u \cos^2 u \\ &= 125 \sin u \end{aligned}$$

$$\iint_S \vec{F} \cdot d\vec{A} = \int_0^{2\pi} \int_0^\pi 125 \sin u \, du \, dv$$

(b) S is closed. Let apply the divergence thm :

$$\operatorname{div} \vec{F} = [1+1+1] = 3$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{A} &= \iiint_T 3 \, dV = 3 \cdot \text{Volume}(T) = 3(2 \cdot 6 \cdot 2) \\ &= 72 \end{aligned}$$

Problem 7:

$$(a) \quad \vec{F} = [xy, 2y^2] \quad \vec{r}(t) = [\cos t, \sin t] \quad 0 \leq t \leq \frac{\pi}{2}$$

$$\vec{r}'(t) = [-\sin t, \cos t]$$

$$\vec{F}(\vec{r}(t)) = [\cos t \sin t, 2 \sin^2 t]$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = -\cos t \sin^2 t + 2 \cos t \sin^2 t = \cos t \sin^2 t$$

$$\text{Work} = \int_0^{\pi/2} \cos t \sin^2 t \, dt$$

$$(b) \quad \delta(x, y, z) = 3\sqrt{5+x} \quad \vec{r}(t) = [t, 2t, \frac{2}{3}t^{3/2}] \quad 0 \leq t \leq 2$$

$$\vec{r}'(t) = [1, 2, t^{1/2}]$$

$$\delta(\vec{r}(t)) = 3\sqrt{5+t} \quad |\vec{r}'(t)| = \sqrt{1+4+t} \underset{t \geq 0}{=} \sqrt{5+t}$$

$$\delta(\vec{r}(t)) \cdot |\vec{r}'(t)| = 3(5+t) \underset{5+t \geq 0}{\uparrow}$$

$$\text{Mass} = \int_0^2 3(5+t) \, dt$$

$$(c) \quad \vec{r}(u, v) = [u, v, u^3] \quad 0 \leq u \leq 1, \quad -2 \leq v \leq 2$$

$$\vec{r}_u = [1, 0, 3u^2] \Rightarrow \vec{r}_u \times \vec{r}_v = [-3u^2, 0, 1]$$

$$\vec{r}_v = [0, 1, 0] \quad |\vec{r}_u \times \vec{r}_v| = \sqrt{9u^4 + 1}$$

$$\text{Surface Area} = \iint_S dA = \int_{-2}^2 \int_0^1 |\vec{r}_u \times \vec{r}_v| \, du \, dv$$

$$\text{Surface Area} = \int_{-2}^2 \int_0^1 \sqrt{9u^4 + 1} \, du \, dv$$

Problem 8: Answer the following true/false questions

T / ☒ F	The line integral of any vector field $\mathbf{F}$ around any closed curve C is zero
T / ☒ F	If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$ for one particular closed path, the $\mathbf{F}$ is path-independent.
☒ T / F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
☒ T / F	If $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} \neq 0$ for some closed path C , then $\mathbf{F}$ is path-dependent.
☒ T / F	If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
☒ T / F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = 0$
☒ T / F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\mathbf{F} = [x, y, z]$ then $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ is positive
T / ☒ F	Let S be a sphere of radius 1 centered at the origin and oriented outward. If $\iint_S \mathbf{F} \cdot \mathbf{n} dA = 0$, then $\mathbf{F} = \mathbf{0}$
T / ☒ F	If $\mathbf{F}$ and $\mathbf{G}$ are vector-fields satisfying $\text{div}(\mathbf{F}) = \text{div}(\mathbf{G})$, then $\mathbf{F} = \mathbf{G}$
☒ T / F	If $\mathbf{F}$ is a vector-field such that $\text{div}(\mathbf{F}) = 1$ and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot \mathbf{n} dA$ is equal to the volume enclosed by S .